As a $\mathbb{Z}$-module, $\mathrm{FN}^{\mathrm{db}}(\Gamma,\mathbb{Z})$ is generated by p_γ, o_γ. For $\gamma>0$ we have $o_\gamma = q_0 - q_\gamma - p_\gamma$, and for $\gamma<0$ we have $o_\gamma = q_\gamma - q_0 - p_0$. The convention $o_0 = -p_0$ is a convention for functions; it does not identify o_0 with the characteristic function of the empty interval $(0,0)$. The convolution products are given by the following cases.

Resolved referee R067 [Error] – Negative interval generators and bounded support. *Original source line 1600.* For $\gamma<0$, the interval $(0,\gamma)$ is empty, so the definition as written cannot yield the claimed generators or quotient: the relation $o_\gamma + p_0$ would even kill the unit p_0. Define $o_\gamma = \mathbf{1}_{(\gamma,0)}$ when $\gamma<0$, retaining $o_\gamma = \mathbf{1}_{(0,\gamma)}$ when $\gamma>0$ and $o_0 = -p_0$. Also define $\mathrm{FN}^{\mathrm{db}}(\Gamma,\mathbb{Z})$ explicitly as the subgroup of functions whose nonzero locus is doubly bounded. These conventions make the subsequent formulas meaningful.

Author resolution. The bounded nonzero-support convention and both signs of the interval are now explicit, including the separate function convention at zero. The literal negative interval cannot supply the intended generator.

For $\beta\neq 0$, put $a=\min(\alpha,\alpha+\beta)$ and $b=\max(\alpha,\alpha+\beta)$. Then $p_\alpha o_\beta$ is the characteristic function of (a,b), and

$$p_\alpha o_\beta = \begin{cases} o_b - o_a - p_a, & 0\le a<b,\\ o_a - o_b - p_b, & a<b\le 0,\\ o_a + o_b + p_0, & a<0<b. \end{cases}$$

For $\beta=0$, $p_\alpha o_0 = -p_\alpha$. Also,

$$o_\alpha o_\beta = \begin{cases} -o_{\alpha+\beta}, & \alpha,\beta\ge 0 \text{ or } \alpha,\beta\le 0,\\ -o_\alpha - o_\beta - p_0, & \alpha<0<\beta \text{ or } \beta<0<\alpha. \end{cases}$$

Resolved referee R068 [Clarification] – Specify which interval identity applies. *Original source line 1605.* After correcting the negative-interval convention, the alternatives can be made unambiguous as follows. Set $a=\min(\alpha,\alpha+\beta)$ and $b=\max(\alpha,\alpha+\beta)$. For $\beta\neq 0$, $p_\alpha o_\beta$ is the characteristic function of (a,b): it is $o_b - o_a - p_a$ if $0\le a<b$, $o_a - o_b - p_b$ if $a<b\le 0$, and $o_a + o_b + p_0$ if $a<0<b$. For $\beta=0$ it equals $-p_\alpha$. Likewise $o_\alpha o_\beta = -o_{\alpha+\beta}$ when $\alpha,\beta\ge 0$ or $\alpha,\beta\le 0$, and equals $-o_\alpha - o_\beta - p_0$ when α and β have opposite strict signs.

Author resolution. The original alternatives are now assigned their sign and endpoint cases, with the zero interval parameter treated separately.

Notation 4.12. The homomorphism

$$\lambda : \mathbf{K}\,\mu\Gamma^{\mathrm{db}}[*] \longrightarrow \mathbb{Z}[\Gamma_{\mathrm{def}}] \oplus X\,\mathrm{FN}^{\mathrm{db}}(\Gamma,\mathbb{Z})[X]$$

is constructed as before (there is only one such homomorphism now since χ_g, χ_b agree on doubly bounded sets).

Notation 4.13. Recall that $\mathrm{RV}^{\circ\circ}_\gamma = \mathrm{rv}(\mathcal{M}_\gamma \smallsetminus 0)$ (Notation 2.5). Let D_{RV} consist of the nonnegative definable γ such that $\gamma^\sharp$ contains an $\mathbb{S}$-definable point. For each $\gamma\in D_{\mathrm{RV}}$, let

$$\boldsymbol{P}_\gamma = [\mathrm{RV}^{\circ\circ} \smallsetminus \mathrm{RV}^{\circ\circ}_\gamma] + [\{t_\gamma\}] - [1] \in \mathbf{K}\,\mathrm{RV}^{\mathrm{db}}[1],$$

where $t_\gamma\in\gamma^\sharp$ is any definable point. It also stands for the corresponding element in $\mathbf{K}\,\mu\mathrm{RV}^{\mathrm{db}}[1]$ (with the constant volume form 0). Note that, although $[\{t_\gamma\}]=[1]$ in $\mathbf{K}\,\mathrm{RV}^{\mathrm{db}}[1]$, $[\{t_\gamma\}]\neq[1]$ in $\mathbf{K}\,\mu\mathrm{RV}^{\mathrm{db}}[1]$ unless $\gamma=0$.

Resolved referee R069 [Error] – The radius must admit a definable RV lift.
Original source line 1621. Not every definable γ admits the definable point t_γ required here. Restrict the definition to nonnegative γ for which $\gamma^\sharp$ contains an $\mathbb{S}$-definable point, equivalently $\gamma \in \mathrm{vrv}(\mathrm{RV}(\mathrm{dcl}\,\mathbb{S}))$. In the later valued-field-generated setting this is the value group of the parameter field, rather than its divisible hull. For example, over $\mathbb{C}((t))$, $\gamma = 1/2$ is definable but there is no definable RV element of that value. Make the same restriction in the generating set of $(\boldsymbol{P}_\Gamma)$. The later specialization to integral radii should refer back to this restriction.

Author resolution. The author approved the restriction to nonnegative radii admitting an RV lift definable over the original parameter base. The set D_{RV} now indexes both the generators and the generated ideal.

Clearly $\boldsymbol{P}_\gamma$ does not depend on the choice of $t_\gamma \in \gamma^\sharp$. The ideal of $\mathbf{K}\,\mu\mathrm{RV}^{\mathrm{db}}[*]$ generated by the elements $\boldsymbol{P}_\gamma$ with $\gamma \in D_{\mathrm{RV}}$ is denoted by $(\boldsymbol{P}_\Gamma)$. A definable Gamma value need not have a definable RV lift, as already noted after Definition 3.13. All choices are made over the original parameter base. The images of $(\boldsymbol{P}_\Gamma)$ under the obvious forgetful homomorphisms

$$\mathbf{K}\,\mu\mathrm{RV}^{\mathrm{db}}[*] \longrightarrow \mathbf{K}\,\mathrm{RV}[*], \quad \mathbf{K}\,\mu\mathrm{RV}^{\mathrm{db}}[*] \longrightarrow \mathbf{K}\,\mu\mathrm{RV}[*]$$

are contained in $(\boldsymbol{P}-1)$, $(\boldsymbol{P})$, respectively.

Resolved referee R070 [Typo] – Closing punctuation.
Original source line 1627. End this sentence with a full stop, not a comma.

Author resolution. The sentence now ends with a full stop.

Remark 4.14. Let $!P$ denote the ideal of $\mathrm{FN}^{\mathrm{db}}(\Gamma,\mathbb{Z})$ generated by the elements $p_\gamma - p_0$, $o_\gamma + p_0$. By the computation in Notation 4.11, $\mathrm{FN}^{\mathrm{db}}(\Gamma,\mathbb{Z})/!P \cong \mathbb{Z}$; the quotient homomorphism is denoted by χ because a simple computation shows that if $\boldsymbol{I} = (I,\mu) \in \mu\Gamma^{\mathrm{db}}[*]$ then indeed $\lambda_{\boldsymbol{I}}/!P = \chi(I)$. Now it is straightforward to check that the assignments

$$p_\gamma X^k \longmapsto [\mathbb{G}_m]^k, \quad o_\gamma X^k \longmapsto -[\mathbb{G}_m]^k \tag{4.7}$$

induce a graded ring homomorphism

$$\psi^{\mathrm{db}} : \mathbb{Z}[\Gamma_{\mathrm{def}}] \oplus X\,\mathrm{FN}^{\mathrm{db}}(\Gamma,\mathbb{Z})[X] \longrightarrow !\mathbf{K}\,\mathrm{RES}[*].$$

In a nutshell, the homomorphism $\psi^{\mathrm{db}} \circ \lambda$ is a "forgetful" map given by $[(I,\mu)] \longmapsto \chi(I)[\mathbb{G}_m]^k$, $(I,\mu) \in \mu\Gamma^{\mathrm{db}}[k]$.

The forgetful homomorphism $\phi : \mathbf{K}\,\mu\mathrm{RES}[*] \longrightarrow !\mathbf{K}\,\mathrm{RES}[*]$ is similar to the one in (4.6).

Proposition 4.15. *There is a graded ring homomorphism*

$$\mu\mathbb{E}^{\mathrm{db}} : \mathbf{K}\,\mu\mathrm{RV}^{\mathrm{db}}[*] \longrightarrow !\mathbf{K}\,\mathrm{RES}[*]$$

such that $(\boldsymbol{P}_\Gamma)$ *vanishes and, for all* $x \in \mathbf{K}\,\mu\mathrm{RES}[*]$, $\mu\mathbb{E}^{\mathrm{db}}(x) = \phi(x)$.

Proof. The homomorphism $\phi \times (\psi^{\mathrm{db}} \circ \lambda)$ is still $\mathbf{K}\,\mu\Gamma^{\mathrm{fin}}[*]$-bilinear. If $\gamma = 0$, the annulus is empty and the value-zero singleton classes $[\{t_0\}]$ and $[1]$ agree, so $\boldsymbol{P}_0 = 0$. For an indexed radius $\gamma > 0$, we have

$$(\mu\Psi^{\mathrm{db}})^{-1}(\boldsymbol{P}_\gamma) = [(0,\gamma)] + [\{\gamma\}] + [\{t_\gamma\}] - [1].$$

Its image under $\phi \otimes (\psi^{\mathrm{db}} \circ \lambda)$ is

$$\chi((0,\gamma))[\mathbb{G}_m] + \chi(\{\gamma\})[\mathbb{G}_m] + [1] - [1] = -[\mathbb{G}_m] + [\mathbb{G}_m] = 0.$$

The open-interval-plus-endpoint expansion is used only for $\gamma > 0$.